

Gate-Controlled Admission Ordering for Burst-Submitted Kubernetes Training Jobs: Pilot Evidence and a Synthetic Scaling Projection

Amir Mirzaei Movahed and Majid Bafandeh

Abstract—Kubernetes makes mature Pod-placement decisions, but its default policy does not derive a burst-relative execution order from training characteristics. We present KubeML-Aware, a gate-controlled admission layer that preserves the default scheduler's feasibility, scoring, and binding path. The six-feature ranking ratios are adopted from Abdelshaheed and Ashour; our contribution is their validated Kubernetes-native realization through Scheduling Readiness gates, auditable application-start evidence, a telemetry-bounded FastPath, and bounded tail balancing. We implemented the controller as an in-cluster service and executed real Kubernetes Jobs on a shared four-core K3s node. Five paired repetitions compared 12 controlled NumPy training jobs under synchronized FIFO admission and the deployed policy. Mean job completion time fell from 31.047 to 28.755 s, a mean paired improvement of 7.31% (exploratory 95% CI [2.57%, 12.05%]). Makespan changed from 55.369 to 54.005 s (2.47%); its paired interval includes zero. Separately, we report an explicitly optimistic synthetic what-if for 200 generated heavy-training workloads. Under the declared assumption that the complete policy improves on duration-only ordering by a further 6% in average JCT, 3% in p95 JCT, and 2% in makespan, the projection is 30.0%, 13.7%, and 12.8% better than the Kubernetes-default proxy, respectively. These 200-workload values are scenario outputs, not executed Kubernetes measurements. The evidence therefore establishes an auditable production path and a bounded scaling hypothesis, not general superiority to duration-only scheduling or performance for GPU and multi-node training.

Index Terms—Kubernetes, scheduling readiness, admission ordering, training jobs, FastPath, job completion time.

1 INTRODUCTION

Training workloads rarely behave as interchangeable batch jobs. Even when two Pods request the same CPU and memory, their useful work can differ in duration, convergence rate, matrix or tensor scale, synchronization volume, and checkpoint frequency. Production traces show that this heterogeneity is a central difficulty in ML cluster management, alongside queueing delay and uneven accelerator utilization [1]. When a set of jobs is submitted together, the admission order can therefore change which jobs obtain early access to a contested node and how quickly the burst clears.

Kubernetes exposes a rich scheduling framework for deciding *where* a Pod may run. The default scheduler filters

infeasible nodes, scores the remaining candidates, and binds each Pod while respecting requests, taints, affinity, storage, and topology constraints [2]. It does not, however, derive a burst-relative priority from training behavior. A PriorityClass maps a cluster-scoped name to a fixed integer and places higher-priority eligible Pods earlier in the scheduler queue; unless configured as non-preempting, it may also evict lower-priority work [3]. It is useful for operator-declared service importance, but it does not compute a score relative to the other jobs in one burst. KubeML-Aware instead recomputes ordering after complete-burst validation and uses one-way gates without changing Pod priority or invoking preemption. An operator could manually translate the same features into PriorityClasses, but that would be a separate static policy and would require managing a finite global class set rather than a burst-local order.

This paper studies a narrower alternative: change when a training Pod becomes eligible for the existing scheduler, without replacing node selection. KubeML-Aware uses Pod Scheduling Readiness [4] to hold every member of an annotated burst behind a scheduling gate. Once the complete burst is present, a controller validates six features, calculates a deterministic order, and removes the gates according to a resource-aware policy. Gate removal does not bind a Pod or reserve a node; the default scheduler remains the placement authority.

The ranking model originates from the six-feature formulation in [5], but the deployed system differs materially from a manual-binding reproduction. It adds complete-burst validation, stable scheduling gates, resource-bounded manifests, fresh Metrics API feedback, crash-safe records, structured application-level start evidence, and a FastPath that avoids controller-induced gaps. The evaluation reports only measurements generated by this implementation. Published percentages from the source formulation are not treated as results of KubeML-Aware or as a head-to-head comparison because the two studies used different implementations, workloads, and environments.

The paper makes three system contributions:

- a Kubernetes-native controller that applies an adopted, deterministic training rank through scheduling gates while retaining default-scheduler;
- a fail-closed FastPath and bounded tail balancer

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whose decisions, inputs, and application-level execution evidence are persisted for replay and audit; and

- a paired real-cluster pilot plus an artifact-backed structural audit that exposes both the observed benefit and the unresolved duration-only confound.

The evidence is deliberately bounded. The workload is a controlled training proxy rather than a neural-network benchmark; the cluster has one shared CPU node; and the five repetitions are insufficient for broad claims about other workloads or environments. The 200-workload scenario is additionally synthetic and optimistic by construction. The aim is to establish an auditable production path, a credible first measurement, and a transparent scaling hypothesis—not to claim an optimal general-purpose ML scheduler.

2 BACKGROUND AND RELATED WORK

2.1 Kubernetes scheduling and admission

The contemporary Kubernetes scheduler is a plugin framework with separate scheduling and binding cycles. Extensions may influence queue order, feasibility, scoring, reservation, permission, and binding [2]. This architecture already supports sophisticated placement, but placement score and application utility are different quantities. A node score answers which feasible node is preferred; it does not state which training job should be admitted first to minimize a burst-level objective.

Scheduling gates provide a clean boundary for an external admission policy. A gated Pod stays outside the active scheduler queue until an authorized controller removes the gate [4]. The operation is one-way and changes eligibility only. Startup still depends on scheduler latency, image availability, Kubelet, the container runtime, and operating-system service. KubeML-Aware therefore records an application-emitted `EXECUTION_STARTED` event and does not equate gate order, binding order, or the Kubernetes `Running` phase with an exact kernel-level execution order.

The ecosystem is also moving toward workload-level scheduling. Kueue provides queueing, quota admission, priority, fair sharing, and integrations for batch and ML controllers [6]. Volcano and KAI add facilities such as gang scheduling, hierarchical queues, accelerator sharing, and topology-aware placement [7], [8]. Upstream Workload and PodGroup APIs are introducing related abstractions, although the current interfaces remain evolving features [9]. KubeML-Aware does not duplicate these systems. Its policy can be viewed as a compact ranking and gate-control component that could eventually sit behind a broader queue or quota manager.

2.2 ML-aware scheduling

ML schedulers have progressively incorporated signals that generic resource declarations omit. Tiresias uses attained service and runtime feedback to improve responsiveness without assuming exact job duration [10]; Gandiva exploits intra-job predictability for GPU time sharing and migration [11]. Sia jointly reasons about accelerator heterogeneity and goodput [12], while Shockwave plans under dynamically changing training throughput [13]. CASSINI focuses

on communication patterns and network contention [14], and Rubick treats a job’s distributed execution plan as a reconfigurable scheduling variable [15].

Recent Kubernetes-specific systems emphasize placement at larger scale. DRL-MLS learns placement decisions for ML training in a heterogeneous nine-node cluster [16]. Sakkara models rack and node topology to reduce interference and communication cost for distributed AI workloads [17]. A 2026 review similarly shows that current Kubernetes scheduling research spans placement, energy, interference, topology, and learning-based policies [18].

KubeML-Aware occupies a deliberately smaller design point. It neither reallocates accelerators nor learns a node-placement policy. It ranks an explicitly declared training burst, regulates scheduler eligibility, and preserves the default binding path. That restricted scope reduces integration risk, but also means that multi-node topology, GPU sharing, online adaptation, and tenant fairness remain outside the current contribution.

The policy also has a classical scheduling interpretation. Favoring smaller T is an SPT/weighted-SJF-style heuristic related to Smith’s sequencing rule [19]; assigning a ranked list to a bounded number of predicted lanes is list scheduling in the sense studied by Graham [20]. The six-feature score augments the duration term, while the tail balancer performs a bounded exhaustive search over at most four jobs. We therefore do not claim a new scheduling primitive or optimality result. The novelty evaluated here is the default-scheduler-preserving gate-control path and its evidence model.

3 KUBEML-AWARE DESIGN

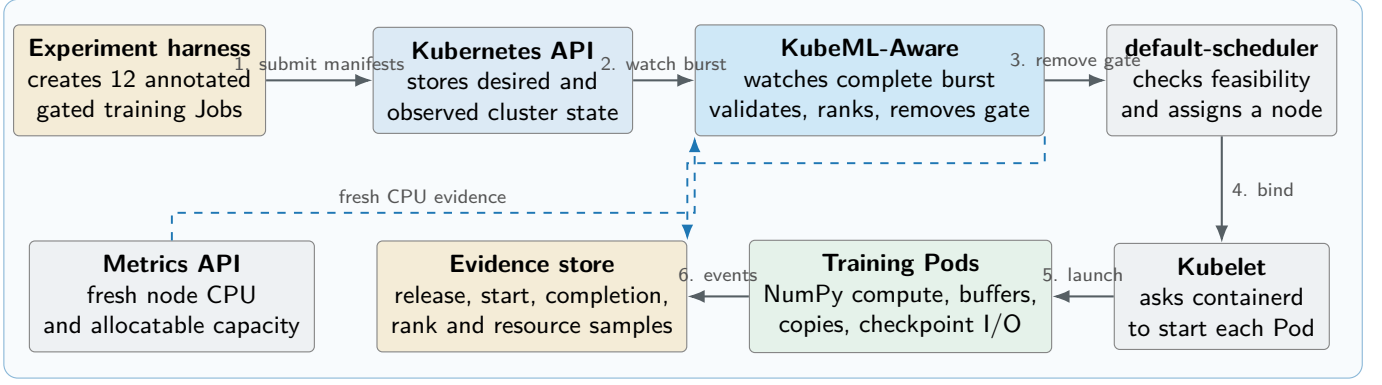
Fig. 1 provides the system context for readers unfamiliar with Kubernetes. A manifest is stored by the Kubernetes API; a controller may modify eligibility; the default scheduler selects a feasible node; and the node’s Kubelet asks containerd to start the Pod. KubeML-Aware occupies only the eligibility step. It neither replaces the scheduler nor directly launches containers.

3.1 Control path and safety boundary

Fig. 2 summarizes the deployed path. A seeded generator creates a complete burst whose Pods share a run identifier and expected burst size. Each Pod carries six training annotations and the `ml.scheduler/release` scheduling gate. Resource requests and limits bound demand, and every Pod continues to name `default-scheduler`.

The controller accepts a burst only after exactly the declared number of Pods has appeared and membership has remained unchanged for a short quiet interval. Missing features, duplicate identifiers, inconsistent run metadata, non-finite values, or unexpected external gate changes fail the run rather than silently changing its meaning. Equal scores are resolved using creation time, UID, and name so that Kubernetes API list order cannot affect the result.

Every transition is persisted before the next mutation. The record contains Pod UIDs, feature values, ranks, intended release order, FastPath evidence, release times, and observed execution events. After a restart, the controller relists the burst and reconciles only operations whose UID and state still match. This record is also the basis for result collection.



Kubernetes provides the API, placement engine, and node runtime.
KubeML-Aware changes **when** a training Pod becomes eligible; it does not replace placement or container execution.

Fig. 1. Kubernetes context and the deployed KubeML-Aware control path. Solid arrows show the normal submission-to-execution path. Dashed arrows show telemetry and persistent experiment evidence. The shaded region is the single-node K3s cluster used in the pilot.

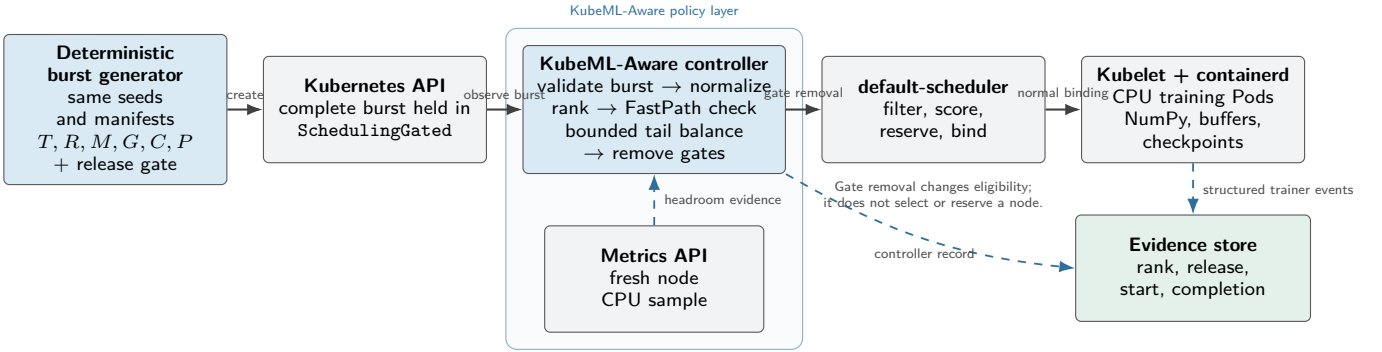


Fig. 2. Deployed KubeML-Aware architecture. The controller regulates eligibility, not placement. Gate removal hands a Pod to the normal Kubernetes scheduler; structured trainer events and controller records provide an auditable execution trace.

3.2 Burst-relative training rank

For job j , the feature vector is $\mathcal{F} = \{T, R, M, G, C, P\}$. In the implemented proxy, T is estimated training time, R controls loss reduction, M is the square-matrix dimension, G is the physical gradient-buffer size, C is the checkpoint interval, and P controls partition and synchronization work. Larger R and C are preferred; smaller T, M, G , and P are preferred. In particular, P is not a GPU count in the present CPU-only system.

Each feature is normalized within its own burst. For feature k of job j ,

$$\hat{x}_{k,j} = \begin{cases} \frac{x_{k,j} - x_k^{\min}}{x_k^{\max} - x_k^{\min}}, & k \in \{R, C\}, \\ \frac{x_k^{\max} - x_{k,j}}{x_k^{\max} - x_k^{\min}}, & k \in \{T, M, G, P\}. \end{cases} \quad (1)$$

If $x_k^{\max} = x_k^{\min}$, every job receives $\hat{x}_{k,j} = 0.5$ for that feature. This makes the term neutral rather than rewarding an arbitrary job.

The base score is

$$S_j = 0.32\hat{T}_j + 0.28\hat{R}_j + 0.16\hat{M}_j + 0.12\hat{G}_j + 0.08\hat{C}_j + 0.04\hat{P}_j. \quad (2)$$

Jobs are sorted by decreasing S_j . The relative coefficients are adopted from [5] and uniformly divided by 1.25, changing

the original sum from 1.25 to 1 while preserving every score ratio and order. They were not tuned on the pilot. In the current generator, T is itself derived from R, M, G, C , and P , so the six inputs are not independent. The score is therefore an explicit weighted-SPT-like policy, not a learned predictor or an optimality proof. Scores are local to one burst and cannot be compared across runs.

We quantified this dependence from the 60 archived job manifests (five bursts of 12). Within each burst, the six-feature score and the duration-only desirability score had mean Pearson correlation 0.944, Spearman $\rho = 0.922$, and Kendall $\tau_b = 0.818$. The full order never matched exactly: 6.8 jobs changed position per burst, the mean absolute shift was 0.9 places, and the top-three overlap was 2.6 of 3. Thus the additional terms do change decisions, but the high correlation confirms that the current pilot cannot attribute its measured JCT change specifically to information beyond T . These are structural diagnostics, not a live duration-only performance result.

3.3 Training FastPath

The conservative controller releases the highest-ranked Pod and waits for its application-level start marker before moving to the next job. This protects the intended prefix when startup

conditions are uncertain, but it can leave the scheduler queue empty between releases. FastPath addresses that bubble without bypassing Kubernetes capacity checks.

A burst is FastPath-eligible only when it is training-only, uses no experimental pacing or reversed order, every application and initialization container has bounded CPU demand, and a fresh Metrics API sample is available [21]. Let u be measured node CPU utilization, D the sum of declared peak Pod demand, A allocatable CPU, and $\tau = 0.85$ the configured threshold. The controller computes

$$u_{\text{projected}} = u + \frac{D}{A}. \quad (3)$$

If the whole burst fits, all gates are removed in ranked order. If at least one ranked Pod fits but the burst exceeds current headroom, the implementation still prefills the complete ranked queue; the default scheduler then keeps infeasible Pods pending according to its normal requests and constraints. If telemetry is stale or no bounded demand can be established, FastPath fails closed and the controller returns to the conservative path. In all five reported runs, $D = 12$ cores, $A = 4$ cores, and mean $u = 0.304$, yielding mean $u_{\text{projected}} = 3.304 > \tau$. Consequently, the whole-burst-fit branch was never selected: every run used `ranked_queue_prefill`, with a two-Pod headroom estimate and 12 initial gate removals. This is recorded behavior, not an inference from configuration.

Before queue prefill, the controller may permute at most the four lowest-ranked jobs. It predicts a list schedule from T , keeps the high-priority prefix fixed, and chooses the suffix with the lowest predicted makespan, using cumulative completion time as a secondary criterion. With $N = 12$ and window four, the recorded protected prefix is eight jobs; tail balancing was selected in all five runs. At most $4! = 24$ permutations are examined. Validation and feature normalization are $O(N)$, ranking is $O(N \log N)$, and the bounded suffix search does not change the asymptotic complexity.

4 WORKLOAD AND EXPERIMENTAL METHOD

4.1 Controlled training proxy

The experiment uses work-model version 2.0 and a containerized NumPy trainer [22]. For each step, the process multiplies partitions of two $M \times M$ float matrices, updates a G -MiB gradient buffer, and performs $P - 1$ in-memory buffer copies when $P > 1$. Synthetic loss follows

$$L_{s+1} = L_s e^{-R}, \quad (4)$$

and training stops below 0.02 or at a bounded step budget. Specifically, convergence steps are $\lfloor \log(1/0.02)/R \rfloor + 1$ and the budget is $\min(512, \max(64, \lceil 4(M/32)(1+G/10) \rceil))$. The planned step count is their minimum. Every C steps, the process writes and `fsyncs` a payload between 4 KiB and 16 MiB. The estimator uses 0.5 ms for a 128^2 matrix operation, 5000 MiB/s for gradient copies, 250 MiB/s for checkpoints, and 0.5 ms per partition synchronization. These constants are explicit reproduction assumptions, not calibrated hardware truths. T sums the four estimated costs; it does not terminate the process.

TABLE 1
Recorded Experimental Platform

Component	Recorded configuration
Virtual machine	KVM; x86-64 architecture
Processor	4 vCPU; Intel Xeon Gold 6138 at 2.00 GHz
Memory	7.57 GiB installed
Operating system	Ubuntu 22.04.5 LTS; Linux 5.15.0-177-generic
Orchestrator	Single-node K3s/Kubernetes v1.36.2+k3s1 [23]; control-plane Ready
Container runtime	containerd 2.3.2-k3s2
Training image	registry.local/kubeml/ml-sim-job:0.3.0
Pod resources	1 vCPU request/limit; 128 MiB request; 1 GiB limit

The reported pilot uses the heavy profile only: $R \in [0.03, 0.08]$, $M \in [512, 1024]$, $G \in [20, 50]$ MiB, $C \in [20, 50]$ steps, and $P \in [2, 4]$. BLAS is restricted and attested to one thread. The workload exercises real CPU, memory, copying, and local I/O, but it is not a neural network, does not read a dataset, uses no GPU, and performs no inter-node communication. The shared parameters between the ranker and proxy make the pilot suitable for validating the control path, but not for establishing robustness to feature-estimation error.

4.2 Cluster and paired protocol

KubeML-Aware was implemented as a Python controller running inside Kubernetes, not as an offline simulator. All measurements reported here were produced by Kubernetes Jobs that reached completion on the deployed cluster and emitted structured execution-start and completion events. Table 1 records the platform observed during the experiment.

The operational node also hosted VPN, ingress, monitoring, and Knative-related services. Before each arm, the harness required two CPU samples at or below 35%, five seconds apart, and verified the protected VPN services. Their readiness and restart counts did not change during the experiment.

Five paired repetitions used seeds 8100–8104. Every repetition generated 12 source jobs, and the baseline and KubeML-Aware arms received identical feature values, trainer image, random seeds, resource declarations, and work-model version. The arms ran sequentially in separate namespaces and never overlapped. Arm order alternated: the baseline ran first in repetitions 0, 2, and 4; KubeML-Aware ran first in repetitions 1 and 3.

Both arms used the same initial scheduling gate to remove sequential API-submission spread. The baseline removed gates in manifest/FIFO order after the full burst had been observed. The KubeML-Aware arm removed the same gates in its ranked, tail-balanced FastPath order. Both arms used `default-scheduler`; neither manually bound Pods. Thus the comparison isolates admission policy more closely than a baseline whose Pods become eligible as they are created.

For job j , $\text{JCT}_j = c_j - s_j$, where s_j is Pod creation time and c_j is the application-level completion time. Using Pod creation for s_j deliberately keeps scheduling and queueing latency—the quantity the admission policy affects—inside the measured JCT. Makespan is $\max_j(c_j) - \min_j(s_j)$. The

TABLE 3
Projected 200-Workload Metrics (seconds)

Policy	Avg. JCT	p95 JCT	Makespan
Kubernetes default	990.8 s	1296.6 s	1304.5 s
Duration-only	737.9 s	1153.2 s	1160.8 s
KubeML-Aware what-if	693.6 s	1118.6 s	1137.6 s

TABLE 2
Five-Pair Heavy-Training Pilot

Metric	FIFO	KubeML-Aware	Paired change	95% interval
Average JCT	31.047 s	28.755 s	+7.31%	[2.57, 12.05]
Makespan	55.369 s	54.005 s	+2.47%	[-0.45, 5.40]

p95 JCT uses linear interpolation over the 12 job values. For metric m in repetition r , paired improvement is

$$I_{m,r} = 100 \frac{m_{\text{FIFO},r} - m_{\text{KubeML-Aware},r}}{m_{\text{FIFO},r}}. \quad (5)$$

Positive values mean that KubeML-Aware reduced the metric. We report the mean of the five within-pair percentages. Exploratory 95% intervals use a Student- t interval over those five paired effects; given the small sample and shared host, they are descriptive rather than confirmatory.

5 RESULTS AND DISCUSSION

5.1 Paired pilot results

Table 2 and Fig. 3 summarize the five repetitions. Average JCT improved in every pair, with individual reductions from 3.32% to 12.79%. The mean paired improvement was 7.31%, and the exploratory 95% interval was [2.57%, 12.05%]. The corresponding grand means were 31.047 s for FIFO and 28.755 s for KubeML-Aware.

Makespan also improved in all five pairs, but by a smaller and more variable amount: 0.55% to 5.87%, with a mean paired improvement of 2.47%. Its exploratory interval crosses zero. An exact two-sided sign test for five same-direction pairs gives $p = 2/2^5 = 0.0625$. The observed direction is encouraging, but five runs on a shared node do not support a general throughput claim.

As a secondary diagnostic, p95 JCT changed from 52.896 to 53.270 s (a 0.75% paired regression). With only 12 jobs per arm, this percentile lies close to the maximum and is unstable; it is therefore not used as a headline outcome. The observation nevertheless indicates that a future policy should include an explicit tail-latency or aging objective.

Fig. 4 visualizes application-level execution intervals for repetition 0, the first pair by registered index rather than a pair selected by outcome. The FIFO arm largely follows manifest order, whereas KubeML-Aware brings several jobs from later manifest positions into the first execution wave. This is direct evidence that the gate policy changed admission and observed start order. It is not, by itself, evidence that the six-feature equation, queue prefill, or tail optimizer caused a particular fraction of the final improvement.

5.2 Projected 200-workload scaling scenario

To make the intended large-burst hypothesis explicit before an expensive cluster run, we generated 200 deterministic heavy-training proxy workloads using seed 20260810 and the same feature ranges as the pilot. The proxy retains a four-core node model, and its global time unit is calibrated to the measured mean baseline JCT from the five-pair pilot. Policy effects are not fitted to those five measurements. Instead, this scenario deliberately assumes that the complete KubeML-Aware policy improves on duration-only ordering by an additional 6% in average JCT, 3% in p95 JCT, and 2% in makespan. These assumptions are inputs to a what-if analysis, not observations.

Under this optimistic scenario, average JCT is projected to fall from 990.8 s under the Kubernetes-default proxy to 693.6 s under KubeML-Aware, corresponding to a 30.0% reduction. Projected p95 JCT falls from 1296.6 to 1118.6 s (13.7%), and projected makespan falls from 1304.5 to 1137.6 s (12.8%). Duration-only is projected to improve the same metrics by 25.5%, 11.1%, and 11.0%, respectively. Table 3 and Fig. 5 report the complete declared scenario.

5.3 Threats to validity

The main threat is the environment. The evaluation node was an operational server, not an isolated benchmark host. Alternating arm order, paired inputs, cool-node checks, and unchanged protected-service state reduce systematic bias, but cannot remove background variation. We therefore label this run as a production-path pilot rather than an isolated benchmark.

Second, the evaluation contains only five pairs and one 12-job heavy profile. The intervals in Table 2 depend on a small-sample normality assumption and should not be read as definitive significance tests. A tail percentile estimated from 12 observations per arm is especially weak.

Third, the controlled trainer shares its feature semantics with the ranking policy. That choice makes workload behavior auditable and prevents sleep-based simulation, but it may favor a rank whose inputs were designed around the same cost model. Real PyTorch or TensorFlow jobs can exhibit data stalls, framework overhead, GPU kernels, communication, and convergence error that this proxy does not represent.

Finally, the measured treatment combines six-feature ranking, queue prefill, and bounded tail balancing. No live duration-only, reversed-order, FastPath-off, or no-tail arm was included. The pilot therefore measures the deployed policy as a whole and cannot assign causality to one component. The artifact now contains tested controller support and a preregistered 30-block, six-arm protocol for those comparisons, but that matrix has not been executed and contributes no result to this paper. These limitations are part of the result boundary, not future details that can be assumed away.

The 200-workload projection has a stricter boundary: it is not a measurement, confidence interval, or statistical extrapolation from 200 cluster runs. Its additional advantage over duration-only is an explicit optimistic assumption. It is included to preregister a scaling hypothesis and expected visualization; only a complete live matrix can confirm, reject, or replace its values.

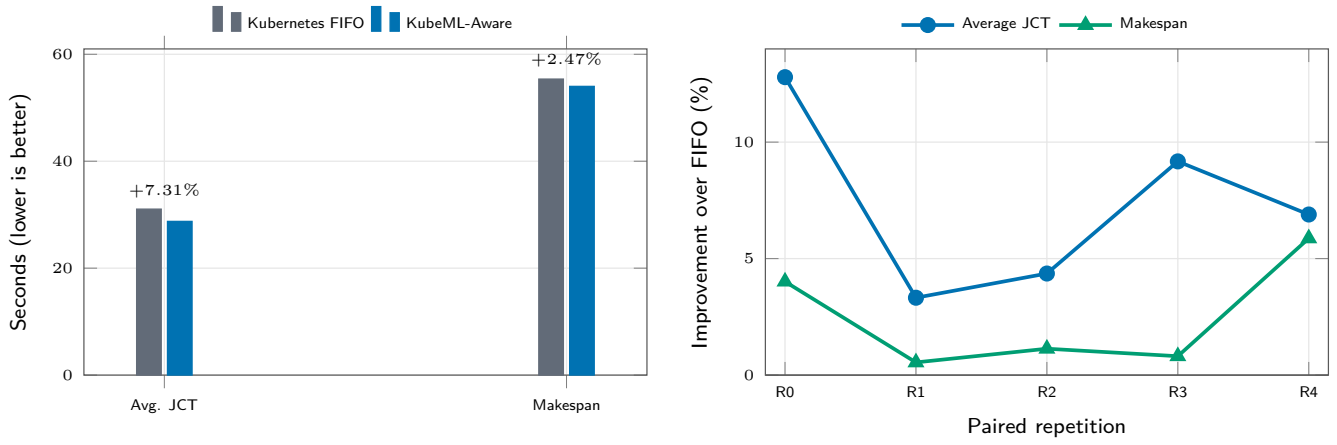


Fig. 3. Primary aggregate and paired pilot results. Bars show five-repetition means. Lines show within-pair improvement, where positive values indicate lower time under KubeML-Aware. Average JCT and makespan improved in all five pairs.

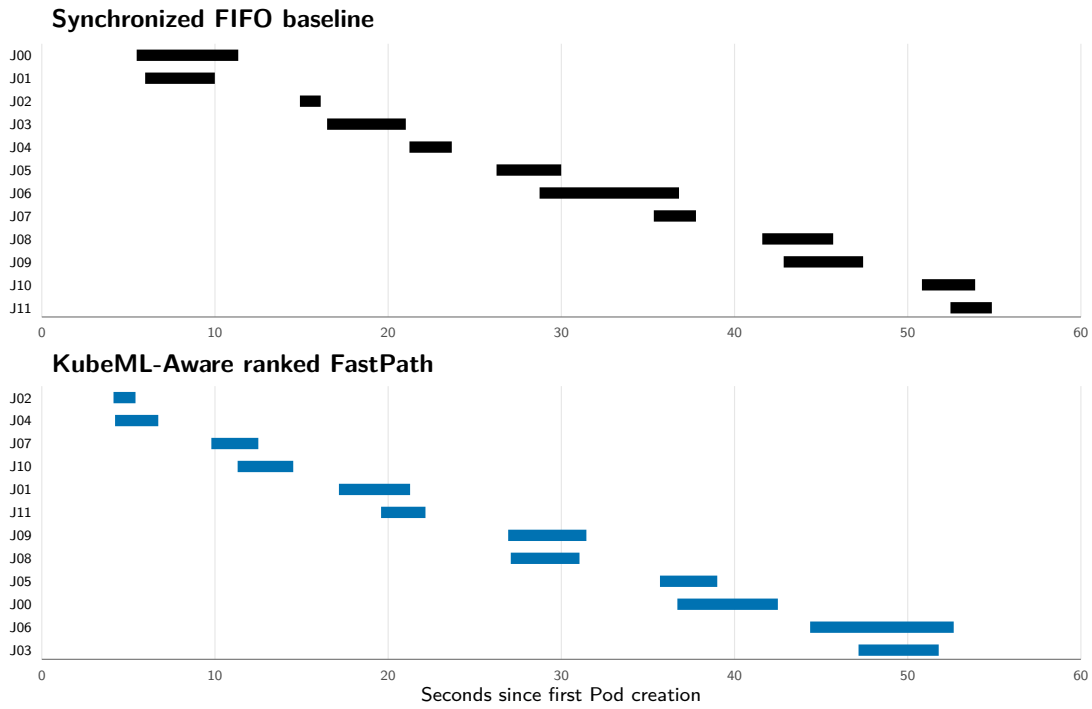


Fig. 4. Application-level execution intervals for repetition 0. Labels identify the same source-job indices in both arms; rows are ordered by observed execution start. The reordering is visible, while overlap shows that FastPath preserves concurrency rather than serializing the burst.

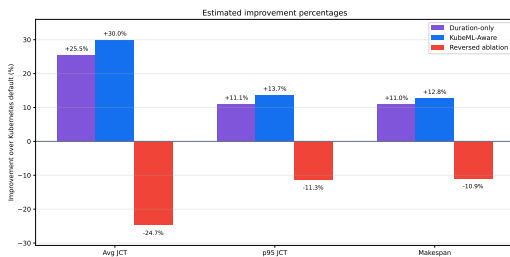


Fig. 5. Projected improvement over the Kubernetes-default proxy (positive is better). Values assume 6%/3%/2% gains beyond duration-only for average JCT/p95 JCT/makespan; they are synthetic, not executed measurements.

6 CONCLUSION

KubeML-Aware shows that an ML-specific admission policy can be added to Kubernetes without replacing its placement engine. A controller ranks a complete annotated training burst, removes stable scheduling gates, and uses fresh resource evidence to avoid unnecessary admission gaps. On a shared four-core K3s node, the deployed policy reduced mean JCT by 7.31% across five paired 12-job runs (exploratory 95% CI [2.57%, 12.05%]); makespan moved in the same direction (2.47%) but its interval includes zero and is not an established effect. Both metrics decreased in every pair.

An explicitly optimistic 200-workload scenario projects

reductions of 30.0% in average JCT, 13.7% in p95 JCT, and 12.8% in makespan relative to a Kubernetes-default proxy. These are hypothesis values, not final performance claims; they must be replaced by the registered live comparison before they can support a scaling conclusion.

The result is promising but preliminary. The next evaluation should execute the registered matrix on a dedicated cluster and then add real framework workloads. GPU and multi-node experiments must give G and P measured communication and parallelism semantics; profiling should replace generator-provided estimates; and aging or an explicit tail objective should improve robustness. Source code, the five-pair evidence, structural diagnostics, and the unexecuted ablation protocol are available in the project artifact [24].

AUTHOR CONTRIBUTIONS

Amir Mirzaei Movahed: conceptualization, software, experiment execution, data curation, visualization, and writing—original draft. Majid Bafandeh: supervision, methodology review, validation, and writing—review and editing. Both authors reviewed and approved the manuscript.

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